

CS634 Final Term Project Report

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Course: CS634 – Data Mining

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Project Title: Binary Classification of Heart Disease Using Random Forest, LSTM, and SVM

1. Introduction

This project focuses on developing and evaluating binary classification models for predicting the likelihood of heart disease using structured clinical data. Binary classification plays an essential role in medical applications because it enables intelligent decision-making, early detection of diseases, and risk stratification based on patient features.

The goal of this work is to compare three machine learning approaches:

1. Random Forest (mandatory ensemble classifier)
2. LSTM (a deep learning model capable of modeling sequential patterns)
3. SVM (a classical ML algorithm known for high-margin decision boundaries)

We evaluate these models using 10-fold cross-validation and a rich set of performance metrics including Accuracy, Precision, Recall, F1-score, Specificity, TSS, HSS, ROC, and AUC. Screenshots, confusion matrices, and ROC curves are included in the Appendix.

2. Dataset

Dataset Name: Heart Disease Dataset

Source: Public repository (Kaggle / UCI versions)

Description:

- Rows: 303
- Features: 13 predictor variables
- Target Variable: target (0 = No Heart Disease, 1 = Heart Disease)
- Type: Binary classification

3. Algorithm overview

3.1 Random Forest

Random Forest is an ensemble of decision trees trained using bagging. Each tree receives a bootstrap sample and randomly selected features, decreasing variance and improving generalization.

3.2 LSTM (Long Short-Term Memory)

LSTM networks contain memory cells and gating mechanisms allowing them to learn long-range interactions. Although not naturally suited for tabular data, an LSTM is included to satisfy the deep learning requirement.

3.3 SVM (Support Vector Machine)

SVM seeks the separating hyperplane with the maximum margin. The RBF kernel captures nonlinear patterns effectively.

4. Implementation

Programming Language: Python

Environment: Jupyter Notebook

Libraries Used: numpy, pandas, sklearn, tensorflow/keras, seaborn, matplotlib

Process:

- Loaded and explored dataset
- Standardized numerical features
- 10-fold cross-validation
- Trained Random Forest, SVM, and LSTM for each fold
- Computed TP, TN, FP, FN and derived evaluation metrics

5. Evaluation Setup

Using 10-fold cross-validation, the following metrics were computed per fold:

Accuracy, Precision, Recall, F1, Specificity, FPR, FNR, TSS, HSS, AUC.

6. Discussion

Random Forest achieved the best overall performance due to its robustness and ability to capture nonlinear interactions.

SVM performed competitively.

LSTM underperformed slightly because the dataset is too small for deep learning.

7. Conclusion

This project demonstrates a complete classification pipeline, with Random Forest emerging as the best-performing model. Cross-validation ensured reliable generalization estimates, and the results reinforce the strengths of ensemble learning for structured medical data.

8. GitHub Link

https://github.com/SathwikSamineni/CS634_Final_Project

